

# Quantifying U.S. Hail Risk: A County-Level Hazard-Exposure-Vulnerability-Loss Framework with GEV-Based Return Period Analysis

## **1. Introduction and Research Question**

Hail is among the costliest convective weather hazards in the United States, generating an estimated 10 to 15 billion dollars in annual insured losses to residential buildings, vehicles, and agricultural crops. Despite this, standard risk assessments, including the Federal Emergency Management Agency's National Risk Index (NRI), apply a single static average loss ratio across all hail sizes and do not account for return-period-conditioned tail risk. This report addresses that gap by developing a full Hazard-Exposure-Vulnerability-Loss pipeline that characterizes hail hazard using extreme value theory, derives diameter-specific damage ratios from observed storm event records, and estimates county-level building and agricultural losses at multiple return periods.

The research question guiding this study is: How does significant hail risk (diameter of at least 1 inch) vary across the U.S. from 1950 to 2024, and what are the estimated county-level building and crop losses across hail size categories using a Hazard-Exposure-Vulnerability-Loss framework with GEV-based return period analysis?

The results directly inform catastrophe reinsurance pricing, crop insurance rate-setting, and future NRI methodology updates. By replacing FEMA's static average loss ratios with diameter-adjusted, return-period-based county-level loss estimates, this report provides a more physically realistic and tail-risk-sensitive characterization of U.S. hail risk than is currently available in the public domain.

The FEMA National Risk Index provides a nationally consistent loss estimation framework but relies on historical averages without extreme value theory extrapolation. This study extends the NRI approach by applying county-level GEV fitting across all data-sufficient U.S. counties, incorporating nonstationarity testing via Mann-Kendall trend analysis, and pairing return-level estimates with diameter-adjusted vulnerability functions to produce loss exceedance curves directly comparable to FEMA benchmarks.

## **2. Data**

### **2.1 NOAA Storm Events Database**

The primary hazard dataset is the NOAA National Centers for Environmental Information Storm Events Database, comprising 75 annual CSV files spanning 1950 to 2024. Each file contains individual severe weather event reports, including event type, county FIPS code, county name, hail diameter in inches, and reported property and crop damage. Records were filtered to significant hail events with diameters of at least 1.0 inch, consistent with NOAA's current severe hail threshold. Events were then grouped into four diameter bins: 1.0 to 2.0 inches, 2.0 to 3.0 inches, 3.0 to 4.0 inches, and 4.0 inches and above, which serve as the vulnerability classification scheme throughout the pipeline (Table 1). Analysis is restricted to the contiguous 48 states; Alaska, Hawaii, Puerto Rico, and other territories were excluded.

The dataset provides county-level, daily event records from 1950 to 2024. Annual maximum hail diameter per county per year was extracted as the block maxima series for GEV fitting. Only counties with at least 10 years of recorded significant hail events were retained, yielding 2,711 qualifying counties shown in Figure 1.

NOAA damage fields are stored as character strings with K/M/B suffixes representing thousands, millions, and billions of dollars respectively. A custom parsing function converted all values to numeric form, with missing damage replaced by zero. Zone-based records and records with FIPS code zero were excluded to retain only true county-level reports.

Hail reporting density increased substantially after 2000 due to expanding spotter networks and social media reporting, potentially introducing an upward bias in apparent hail frequency in the earlier portions of the record. Pre-2000 events are retained per the study period prescribed for this course, but this reporting inhomogeneity is acknowledged as a limitation, particularly for nonstationarity testing. The 1-inch threshold partially mitigates this bias since large-hail spotter coverage is more uniform across decades than coverage of marginal sub-inch events.

## **2.2 FEMA National Risk Index**

County-level exposure and baseline vulnerability data were sourced from the FEMA National Risk Index counties dataset (December 2025 release). The following fields were extracted: Hail Exposure Building Value, Hail Exposure Agriculture Value, Hail Historic Loss Ratio for Buildings, Hail Historic Loss Ratio for Agriculture, and Hail Expected Annual Loss for both building and agricultural categories. The NRI expected annual loss estimates serve as a validation benchmark for modeled losses throughout Section 4.

NRI exposure values reflect 2025 building stock conditions rather than historical values, potentially overestimating losses in earlier years. The NRI historic loss ratios represent averages across all hail sizes, which this study directly improves upon through diameter-specific scaling. County name matching between NOAA Storm Events and NRI required reconciliation for counties with naming discrepancies, including Virginia independent cities, Connecticut planning regions, and counties with spacing differences in multi-word names such as De Kalb versus DeKalb. After reconciliation, all 2,711 counties were successfully matched to NRI exposure data with no counties excluded from loss estimation.

## **3. Methods**

### **3.1 Hazard: Extreme Value Analysis**

Annual maximum hail diameter per county per year was extracted from the filtered storm event records. The Generalized Extreme Value distribution was selected as the hazard model, which establishes the GEV as the limiting distribution for block maxima. The GEV cumulative distribution function is  $F(x, \mu, \sigma, \xi) = e^{-[1+\xi(\frac{x-\mu}{\sigma})]^{-1/\xi}}$ , where  $\mu$  is the location parameter in inches,  $\sigma$  is the scale parameter in inches, and  $\xi$  is the dimensionless shape parameter governing tail behavior. Positive  $\xi$  indicates a heavy-tailed Fréchet distribution; negative  $\xi$  indicates a

bounded Weibull upper tail; and values near zero correspond to the Gumbel type with a light exponential tail. The alternative to GEV considered was fitting a Generalized Pareto Distribution to exceedances above a threshold, following the peaks-over-threshold approach. The block maxima approach was retained because it avoids the need to specify and justify a threshold parameter and produces more stable fits for county-level records with limited data.

Prior to GEV fitting, each county's annual maximum hail series was tested for monotonic trend using the nonparametric Mann-Kendall test. Counties with statistically significant trends ( $p < 0.05$ ) were classified as nonstationary and are shown spatially in Figure 2. Of 2,711 counties tested, 2,380 were classified as stationary, 42 as nonstationary with increasing trends, and 289 as nonstationary with decreasing trends.

For the 2,380 stationary counties, GEV parameters were estimated using the method of L-moments via the R package *lmom*. All 2,380 stationary counties were successfully fitted using L-moments with no MLE fallback required. L-moments were preferred over Maximum Likelihood Estimation for stationary fitting given the limited sample sizes per county, where MLE can exhibit numerical instability. Return levels at 50, 100, and 500 years were computed analytically from the fitted parameters. Parameter and return level distributions across all stationary counties are shown in Figures 5 and 6, and summary statistics are provided in Table 2.

For the 331 nonstationary counties, a time-varying GEV model was fitted with a linearly evolving location parameter of the form  $\mu(t) = \mu_0 + \mu_1 t$ , where  $t$  is years since the start of each county's record. MLE was used for nonstationary fitting via the R package *extRemes*, as it naturally accommodates time covariates. L-moment estimates served as initial values for the MLE optimizer to ensure convergence. Return levels were evaluated at the end of the observational record to reflect current risk conditions. Of the 331 nonstationary counties, 277 were successfully fitted using the full nonstationary MLE model; 49 fell back to stationary L-moments and 5 to stationary MLE due to convergence failures or physically implausible return levels exceeding 10 inches. Parameter and return level distributions across all nonstationary counties are shown in Figures 7 and 8, with summary statistics in Table 3. The spatial distribution of 100-year return levels across all counties is shown in Figure 4, and the classification of each county into its corresponding damage bin at the 100-year level is shown in Figure 3.

### **3.2 Uncertainty Quantification**

Parametric uncertainty in return level estimates was quantified through nonparametric bootstrap resampling with 1,000 iterations for three representative counties: Oklahoma County, Oklahoma as the stationary example; Pennington County, South Dakota as the nonstationary-increasing example; and Jefferson County, Alabama as the nonstationary-decreasing example. For each iteration, the annual maximum series was resampled with replacement and the GEV was refitted using the appropriate method. The 95% confidence intervals were extracted as the 2.5th and 97.5th percentiles of bootstrap return levels. A physical plausibility cap of 12 inches was applied to bootstrap return level samples to prevent numerical instability from inflating

confidence intervals, consistent with the caps applied during the main GEV fitting pipeline. Bootstrap distributions for all three return periods are shown in Figures 9 through 11, and summary confidence intervals are provided in Table 4.

Bayesian uncertainty quantification was performed using the R package `revdbayes` with a flat, non-informative prior on GEV parameters. Posterior distributions were sampled via Markov Chain Monte Carlo with 20,000 draws for each of the three example counties. Return level credible intervals were derived from the 2.5th and 97.5th percentiles of the posterior predictive distribution. MCMC credible intervals are presented alongside bootstrap confidence intervals for comparison in Table 5, and return level plots with 95% posterior credible bands are shown in Figures 12 through 14. Unlike bootstrap confidence intervals, MCMC credible intervals remained physically plausible without additional capping, suggesting greater robustness for uncertainty quantification in counties with short observational records.

### **3.3 Exposure, Vulnerability, and Loss Estimation**

County-level building and agricultural exposure values were drawn from the FEMA NRI hail exposure fields, representing the portion of each county's building stock and crop acreage within the hail hazard footprint.

Rather than applying FEMA's single average loss ratio uniformly across all hail sizes, this study derives diameter-specific damage scaling factors from observed NOAA storm event damage reports. Mean property and crop damages were computed for each hail size bin and are presented in Table 1. Bin multipliers were then computed as the ratio of each bin's mean damage to the overall mean damage across all significant hail events, yielding a scaling factor that adjusts FEMA's county-specific loss ratio up or down based on the GEV return level bin classification at each return period. Bin boundaries were applied with a tolerance of 0.01 inches to prevent misclassification of return levels falling marginally below integer thresholds due to GEV fitting artifacts. These multipliers are shown in Table 6.

County-level losses were estimated as the product of NRI Exposure, FEMA Loss Ratio, and Bin Multiplier, computed separately for buildings and agriculture at the 50-, 100-, and 500-year return periods. An alternative approach, computing damage ratios by dividing NOAA mean damage directly by NRI exposure, was tested but rejected because the two quantities cancel algebraically, producing identical losses for all counties within the same bin regardless of size. The bin multiplier approach was adopted as the methodologically sound alternative that preserves county-level spatial variability while maintaining physical interpretability.

An aggregate U.S. loss exceedance curve was constructed by summing county-level losses across return periods from 2 to 1,000 years using log-linear interpolation between the three fitted return levels. The resulting curve, shown in Figure 15, expresses total U.S. hail loss as a function of return period and is compared against FEMA's aggregate NRI expected annual loss as a validation benchmark.

## **4. Results and Interpretation**

## 4.1 Hazard Characterization

Of the 522,618 significant hail events recorded nationally from 1950 to 2024, 88.7% fell in the 1.0 to 2.0 inch bin, 9.4% in the 2.0 to 3.0 inch bin, 1.1% in the 3.0 to 4.0 inch bin, and 0.8% in the 4.0 inch and above category (Table 1). Mean property damage increases approximately 71-fold from the smallest to the largest bin, rising from roughly \$39,000 to \$2.8 million per event, while mean crop damage increases approximately 14-fold from approximately \$11,000 to \$149,000. Median damage equals zero across all bins, confirming that extreme value methods are required for rigorous risk characterization rather than simple averaging approaches based on historical means.

Stationary GEV location parameters across fitted counties averaged 1.4 inches across fitted counties, ranging from 1.0 to 2.2 inches, consistent with typical annual maximum hail sizes in the continental United States (Figures 5 and 6, Table 2). Shape parameters were predominantly positive, indicating heavy-tailed Fréchet distributions consistent with Table 2, which shows a mean shape parameter of 0.048. The nonstationary county set showed a distribution of trend slopes centered near zero with a mean of -0.009 inches per year (Figures 7 and 8, Table 3), with the 42 increasing-trend counties exhibiting positive slopes, indicating that hail risk is growing in those locations over the period of record.

Mann-Kendall testing identified 331 of 2,711 counties, representing 12.2% of the total, with statistically significant trends in annual maximum hail size (Figure 2). The predominance of 289 decreasing-trend counties relative to 42 increasing-trend counties is a scientifically noteworthy result that may reflect real climatological patterns in storm track frequency, reporting artifacts associated with the pre-2000 record, or a combination of both. The 100-year GEV return level map (Figure 4) shows pronounced hazard concentration in the southern Great Plains, consistent with the climatological Hail Alley corridor across Texas, Oklahoma, and Kansas, where counties commonly exceed 4 inches at the 100-year return level. The Midwest and Southeast show predominantly 2 to 3 inch return levels, while much of the western United States falls in the 1 to 2 inch range or has insufficient data. The hail size bin classification map (Figure 3) translates these continuous return level estimates into the four policy-relevant damage categories used throughout the loss pipeline.

## 4.2 Uncertainty Quantification

Bootstrap 95% confidence intervals for the 100-year return level span approximately 2.09 inches for Oklahoma County (4.45 to 6.54 inches), 4.54 inches for Pennington County (4.68 to 9.22 inches), and 1.12 inches for Jefferson County (2.81 to 3.92 inches), as shown in Table 4 and Figures 9 through 11. The wider interval for the nonstationary-increasing county relative to the stationary county reflects the additional parameter uncertainty from estimating the trend slope, demonstrating the cost of increased model complexity. The narrower interval for the nonstationary-decreasing county reflects the relative stability of that county's hail record over the observational period.

MCMC 95% credible intervals are consistently wider than the corresponding bootstrap confidence intervals across all three counties and all return periods (Table 5, Figures 12 through 14). For example, the 100-year MCMC credible interval for Oklahoma County spans 4.39 to 8.59 inches compared to the bootstrap interval of 4.45 to 6.54 inches. This pattern is expected: Bayesian credible intervals capture full posterior uncertainty across all GEV parameters simultaneously, whereas bootstrap intervals reflect only sampling variability in the observed series. At the 500-year return period, credible intervals widen substantially for all three counties, appropriately communicating the large epistemic uncertainty inherent in extrapolating 7 to 50 times beyond the length of the observational record.

### **4.3 Loss Estimation**

Total modeled 100-year losses across all 2,711 counties amount to approximately \$10.8 billion, compared to FEMA's aggregate NRI expected annual loss of approximately \$3.0 billion. The modeled 100-year loss exceeding FEMA's annual average by a factor of approximately 3.6 is physically coherent: FEMA's expected annual loss represents an unconditional average across all years including near-zero-loss years, while the 100-year return level captures a rare but severe tail event. The loss exceedance curve (Figure 15) shows aggregate U.S. losses ranging from approximately \$0.8 billion at the 2-year return period to approximately \$14.9 billion at the 1,000-year return period. FEMA's total expected annual loss intersects the exceedance curve at approximately the 10-year return period, suggesting that FEMA's annual average corresponds to a relatively frequent loss scenario under this framework. This finding has direct implications for catastrophe reinsurance pricing, where treaties are commonly structured around 100-year and 250-year return period events rather than annual average losses.

The ten highest 100-year building loss counties are dominated by large Texas metropolitan areas, including Bexar County at approximately \$545 million, Hidalgo County at approximately \$414 million, Dallas County at approximately \$324 million, Maricopa County at approximately \$275 million, and Tarrant County at approximately \$274 million, reflecting the combination of extreme hail hazard and very high building exposure in the region (Table 8). Modeled losses exceed FEMA expected annual loss by factors of 2.4 to 44.5 across the top ten counties, with the largest ratios occurring in counties where FEMA's average loss ratio most severely understates tail risk. Agricultural losses follow a different spatial pattern, concentrating in central California and the Great Plains where crop exposure is highest, with Fresno County leading at \$26.3 million and Nebraska counties dominating the remainder of the top ten (Table 9). The FEMA versus modeled loss comparison map (Figure 19) shows that the majority of fitted counties exceed FEMA's expected annual loss at the 100-year return level, consistent with the aggregate finding.

Texas ranks first nationally by total 100-year hail loss at \$3.6 billion, followed by Colorado at \$670 million and Wisconsin at \$524 million (Figure 20, Table 11). The state ranking map clearly differentiates Hail Alley states from lower-risk New England and Pacific Coast states, providing a spatially intuitive summary of national hail risk that is directly actionable for insurers and state-level emergency management agencies. Spatial maps of building losses (Figure 16),

agricultural losses (Figure 17), and total losses (Figure 18) confirm that the two loss components are geographically concentrated in different regions, with building losses peaking in urban Texas corridors and agricultural losses peaking in the crop-intensive Great Plains.

## **5. Discussion and Limitations**

### **5.1 Risk Context and Decision Implications**

This study demonstrates that GEV-based return period analysis captures substantially higher tail losses than FEMA's static average approach, by a factor of 2.4 to 184 at the county level at the 100-year return period. The finding that FEMA's expected annual loss corresponds to roughly the 10-year return period loss has direct implications for catastrophe reinsurers, who may be anchoring aggregate hail portfolio pricing to near-term average loss expectations rather than tail scenarios. This analysis suggests planning for \$10.8 to \$14.4 billion in aggregate industry losses at the 100 to 500-year return period, substantially above historical average annual loss figures. For crop insurers, agricultural loss concentration in Nebraska and central California indicates that reinsurance pools with high Plains and high-value crop exposure carry elevated tail risk that the current NRI methodology does not fully capture. For FEMA and state emergency managers, the 10-year return period calibration of the NRI expected annual loss suggests the index reflects a relatively frequent scenario rather than a long-horizon tail risk benchmark, and return-period-specific loss estimates would strengthen hazard mitigation grant prioritization.

### **5.2 Limitations and Failure Modes**

Hail reporting density increased substantially after 2000, introducing a potential upward bias in apparent hail frequency in the pre-2000 period of the record. Mann-Kendall nonstationarity tests spanning the full 1950 to 2024 record may therefore conflate reporting changes with true climatic trends, and the predominance of decreasing trend counties should be interpreted cautiously given this artifact.

With 10 to 74 years of data per county under the 10-year minimum threshold, 500-year return level estimates require extrapolation between 7 and 50 times beyond the observational record length. These estimates carry substantial uncertainty and should be interpreted as order-of-magnitude guidance rather than precise predictions, particularly in counties with fewer than 15 years of data where GEV fitting is numerically marginal. The bootstrap and MCMC credible bands shown in Figures 9 through 14 explicitly communicate this uncertainty for the three example counties.

NRI exposure values reflect 2025 building stock conditions, introducing a temporal mismatch that likely overestimates losses for historical storms and may underestimate losses for future events in rapidly growing metropolitan areas. Additionally, the vulnerability function assumes a stationary relationship between hail size and damage fraction over 74 years, ignoring evolution in building codes, roofing materials, and construction practices that have likely changed the hail-to-damage relationship substantially since 1950.

NOAA damage estimates are self-reported and inconsistently filed; many damaging events record zero damage if no formal assessment was conducted. This likely understates actual damage ratios, particularly for the 1 to 2 inch bin where minor claims frequently go unreported. Finally, all analysis is performed at the county level, masking substantial sub-county spatial variability in both hail occurrence and property values. High-resolution grid-based analysis using NEXRAD MESH data would substantially improve loss estimates in densely developed urban counties where hail footprints can be smaller than the county area.



## **References**

FEMA, 2025: National Risk Index for Counties. Federal Emergency Management Agency, accessed 17 March 2025, <https://hazards.fema.gov/nri/data-resources>.

NOAA NCEI, 2024: Storm Events Database (1950-2024). National Centers for Environmental Information, accessed 17 March 2025, <https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/>.

## Appendix

**Table 1.** Summary statistics by hail size (1950-2024)

| Hail Size  | Reports | Property Mean | Property Max | Crop Mean | Crop Max |
|------------|---------|---------------|--------------|-----------|----------|
| 1.0+ in    | 522,618 | \$121.4K      | \$2.3B       | \$15.1K   | \$100M   |
| 1.0-2.0 in | 463,558 | \$39.4K       | \$900M       | \$11K     | \$70M    |
| 2.0-3.0 in | 49,364  | \$508.1K      | \$2.3B       | \$38.6K   | \$80M    |
| 3.0-4.0 in | 5,670   | \$1.6M        | \$1.8B       | \$48.6K   | \$55M    |
| 4.0+ in    | 4,026   | \$2.8M        | \$1.4B       | \$149.4K  | \$100M   |

**Table 2.** Stationary GEV summary statistics (L-moments)

| Statistic | Location ( $\mu$ ) | Scale ( $\sigma$ ) | Shape ( $\xi$ ) | 50 Year Return Level | 100 Year Return Level | 500 Year Return Level |
|-----------|--------------------|--------------------|-----------------|----------------------|-----------------------|-----------------------|
| Mean      | 1.365              | 0.394              | 0.048           | 3.178                | 3.627                 | 4.988                 |
| Median    | 1.354              | 0.407              | 0.065           | 3.078                | 3.495                 | 4.591                 |
| Min       | 1.000              | 0.000              | -2.852          | 1.000                | 1.000                 | 1.000                 |
| Max       | 2.152              | 0.873              | 1.000           | 7.514                | 11.292                | 29.830                |

**Table 3.** Nonstationary GEV summary statistics (MLE)

| Statistic | Trend Slope ( $\mu_1$ ) | Scale ( $\sigma$ ) | Shape ( $\xi$ ) | 50 Year Return Level | 100 Year Return Level | 500 Year Return Level |
|-----------|-------------------------|--------------------|-----------------|----------------------|-----------------------|-----------------------|
| Mean      | -0.009 in/yr            | 0.350              | 0.127           | 2.742                | 3.108                 | 4.280                 |
| Median    | -0.010 in/yr            | 0.356              | 0.065           | 2.641                | 2.965                 | 3.949                 |
| Min       | -0.048 in/yr            | 0.128              | -1.647          | 1.000                | 1.000                 | 1.000                 |
| Max       | 0.023 in/yr             | 0.720              | 4.104           | 7.424                | 10.603                | 24.677                |

**Table 4a.** Bootstrap confidence intervals (50-year return level)

| County             | Type                     | 50 Year Return Level (Est.) | 50 Year Return Level (Lower) | 50 Year Return Level (Upper) |
|--------------------|--------------------------|-----------------------------|------------------------------|------------------------------|
| Oklahoma Co., OK   | Stationary               | 4.862                       | 4.039                        | 5.519                        |
| Pennington Co., SD | Nonstationary Increasing | 5.187                       | 4.334                        | 6.965                        |
| Jefferson Co., AL  | Nonstationary Decreasing | 2.998                       | 2.589                        | 3.380                        |

**Table 4b.** Bootstrap confidence intervals (100-year return level)

| County           | Type       | 100 Year Return Level (Est.) | 100 Year Return Level (Lower) | 100 Year Return Level (Upper) |
|------------------|------------|------------------------------|-------------------------------|-------------------------------|
| Oklahoma Co., OK | Stationary | 5.552                        | 4.454                         | 6.543                         |

|                    |                          |       |       |       |
|--------------------|--------------------------|-------|-------|-------|
| Pennington Co., SD | Nonstationary Increasing | 5.869 | 4.683 | 9.219 |
| Jefferson Co., AL  | Nonstationary Decreasing | 3.278 | 2.806 | 3.921 |

**Table 4c.** Bootstrap confidence intervals (500-year return level)

| County             | Type                     | 500 Year Return Level (Est.) | 500 Year Return Level (Lower) | 500 Year Return Level (Upper) |
|--------------------|--------------------------|------------------------------|-------------------------------|-------------------------------|
| Oklahoma Co., OK   | Stationary               | 7.364                        | 5.299                         | 9.808                         |
| Pennington Co., SD | Nonstationary Increasing | 7.716                        | 5.324                         | 18.202                        |
| Jefferson Co., AL  | Nonstationary Decreasing | 3.898                        | 3.183                         | 5.239                         |

**Table 5a.** Markov Chain Monte Carlo confidence intervals (50-year return level)

| County             | Type                     | 50 Year Return Level (Est.) | 50 Year Return Level (Lower) | 50 Year Return Level (Upper) |
|--------------------|--------------------------|-----------------------------|------------------------------|------------------------------|
| Oklahoma Co., OK   | Stationary               | 4.879                       | 4.043                        | 6.914                        |
| Pennington Co., SD | Nonstationary Increasing | 5.207                       | 4.197                        | 8.018                        |
| Jefferson Co., AL  | Nonstationary Decreasing | 3.209                       | 2.870                        | 4.121                        |

**Table 5b.** Markov Chain Monte Carlo confidence intervals (100-year return level)

| County             | Type                     | 100 Year Return Level (Est.) | 100 Year Return Level (Lower) | 100 Year Return Level (Upper) |
|--------------------|--------------------------|------------------------------|-------------------------------|-------------------------------|
| Oklahoma Co., OK   | Stationary               | 5.525                        | 4.393                         | 8.590                         |
| Pennington Co., SD | Nonstationary Increasing | 6.016                        | 4.565                         | 10.472                        |
| Jefferson Co., AL  | Nonstationary Decreasing | 3.423                        | 2.990                         | 4.678                         |

**Table 5c.** Markov Chain Monte Carlo confidence intervals (500-year return level)

| County           | Type       | 500 Year Return Level (Est.) | 500 Year Return Level (Lower) | 500 Year Return Level (Upper) |
|------------------|------------|------------------------------|-------------------------------|-------------------------------|
| Oklahoma Co., OK | Stationary | 7.169                        | 5.107                         | 14.022                        |

|                    |                          |       |       |        |
|--------------------|--------------------------|-------|-------|--------|
| Pennington Co., SD | Nonstationary Increasing | 8.272 | 5.351 | 19.828 |
| Jefferson Co., AL  | Nonstationary Decreasing | 3.850 | 3.172 | 6.168  |

**Table 6.** Hail size bin damage multipliers (100-year return level)

| Hail Size  | Property Multiplier | Crop Multiplier |
|------------|---------------------|-----------------|
| 1.0-2.0 in | 0.324               | 0.729           |
| 2.0-3.0 in | 4.184               | 2.560           |
| 3.0-4.0 in | 12.896              | 3.220           |
| 4.0+ in    | 23.010              | 9.902           |

**Table 7.** Total loss exceedance curve

| Return Period | Total Loss | Exceedance Probability |
|---------------|------------|------------------------|
| 2 yr          | \$0.8B     | 50%                    |
| 5 yr          | \$1.7B     | 20%                    |
| 10 yr         | \$3.2B     | 10%                    |
| 25 yr         | \$6.1B     | 4%                     |
| 50 yr         | \$8.9B     | 2%                     |
| 100 yr        | \$10.8B    | 1%                     |
| 200 yr        | \$13.1B    | 0.5%                   |
| 500 yr        | \$14.4B    | 0.2%                   |
| 1000 yr       | \$14.9B    | 0.1%                   |

**Table 8.** Top 10 counties ranked by building loss (100-year return level)

| County            | Model      | Hail Size  | Building Loss | Total Loss | FEMA EAL | FEMA Comparison (Total Loss vs FEMA EAL) |
|-------------------|------------|------------|---------------|------------|----------|------------------------------------------|
| Bexar Co., TX     | Stationary | 4.0+ in    | \$544.5M      | \$552.1M   | \$72.3M  | 7.6x                                     |
| Hidalgo Co., TX   | Stationary | 4.0+ in    | \$413.8M      | \$414.6M   | \$9.3M   | 44.5x                                    |
| Dallas Co., TX    | Stationary | 4.0+ in    | \$323.5M      | \$323.5M   | \$127.4M | 2.5x                                     |
| Maricopa Co., AZ  | Stationary | 2.0-3.0 in | \$274.5M      | \$274.5M   | \$24.1M  | 11.4x                                    |
| Tarrant Co., TX   | Stationary | 4.0+ in    | \$273.8M      | \$273.8M   | \$112M   | 2.4x                                     |
| Collin Co., TX    | Stationary | 4.0+ in    | \$241.1M      | \$241.1M   | \$96.4M  | 2.5x                                     |
| Warren Co., KY    | Stationary | 4.0+ in    | \$165.6M      | \$165.6M   | \$22.5M  | 7.4x                                     |
| St. Louis Co., MO | Stationary | 4.0+ in    | \$164.1M      | \$164.1M   | \$37.6M  | 4.4x                                     |

|                 |            |            |          |          |         |       |
|-----------------|------------|------------|----------|----------|---------|-------|
| El Paso Co., TX | Stationary | 2.0-3.0 in | \$146.9M | \$146.9M | \$14.5M | 10.1x |
| Travis Co., TX  | Stationary | 4.0+ in    | \$146.1M | \$146.6M | \$24.9M | 5.9x  |

**Table 9.** Top 10 counties ranked by agricultural loss (100-year return level)

| County           | Model                    | Hail Size  | Agricultural Loss | Total Loss | FEMA EAL | FEMA Comparison (Total Loss vs FEMA EAL) |
|------------------|--------------------------|------------|-------------------|------------|----------|------------------------------------------|
| Fresno Co., CA   | Stationary               | 2.0-3.0 in | \$26.3M           | \$37M      | \$2.7M   | 13.9x                                    |
| Lubbock Co., TX  | Stationary               | 4.0+ in    | \$18.2M           | \$138.2M   | \$34.7M  | 4x                                       |
| Okanogan Co., WA | Stationary               | 2.0-3.0 in | \$15.8M           | \$16.3M    | \$617.1K | 26.4x                                    |
| Dawson Co., NE   | Stationary               | 4.0+ in    | \$10.6M           | \$15M      | \$9M     | 1.7x                                     |
| Clay Co., NE     | Stationary               | 4.0+ in    | \$9.5M            | \$13M      | \$8.5M   | 1.5x                                     |
| York Co., NE     | Stationary               | 4.0+ in    | \$9M              | \$15.8M    | \$9.2M   | 1.7x                                     |
| Adams Co., NE    | Nonstationary Increasing | 4.0+ in    | \$9M              | \$23.7M    | \$11.4M  | 2.1x                                     |
| Buffalo Co., NE  | Stationary               | 4.0+ in    | \$8.9M            | \$27.7M    | \$11.9M  | 2.3x                                     |
| Hamilton Co., NE | Stationary               | 4.0+ in    | \$8.9M            | \$12.3M    | \$7.5M   | 1.6x                                     |
| Webster Co., NE  | Stationary               | 4.0+ in    | \$8.5M            | \$11M      | \$7.3M   | 1.5x                                     |

**Table 10.** Top 10 counties ranked by total loss (100-year return level)

| County           | Model      | Hail Size  | Total Loss | FEMA EAL | FEMA Comparison (Total Loss vs FEMA EAL) |
|------------------|------------|------------|------------|----------|------------------------------------------|
| Bexar Co., TX    | Stationary | 4.0+ in    | \$552.1M   | \$72.3M  | 7.6x                                     |
| Hidalgo Co., TX  | Stationary | 4.0+ in    | \$414.6M   | \$9.3M   | 44.5x                                    |
| Dallas Co., TX   | Stationary | 4.0+ in    | \$323.5M   | \$127.4M | 2.5x                                     |
| Maricopa Co., AZ | Stationary | 2.0-3.0 in | \$274.5M   | \$24.1M  | 11.4x                                    |
| Tarrant Co., TX  | Stationary | 4.0+ in    | \$273.8M   | \$112M   | 2.4x                                     |

|                   |            |            |          |         |       |
|-------------------|------------|------------|----------|---------|-------|
| Collin Co., TX    | Stationary | 4.0+ in    | \$241.1M | \$96.4M | 2.5x  |
| Warren Co., KY    | Stationary | 4.0+ in    | \$165.6M | \$22.5M | 7.4x  |
| St. Louis Co., MO | Stationary | 4.0+ in    | \$164.1M | \$37.6M | 4.4x  |
| El Paso Co., TX   | Stationary | 2.0-3.0 in | \$146.9M | \$14.5M | 10.1x |
| Travis Co., TX    | Stationary | 4.0+ in    | \$146.6M | \$24.9M | 5.9x  |

**Table 11.** Top 10 states ranked by total loss (100-year return level)

| State | Total Counties | Building Loss | Agricultural Loss | Total Loss | FEMA EAL | FEMA Comparison (Total Loss vs FEMA EAL) |
|-------|----------------|---------------|-------------------|------------|----------|------------------------------------------|
| TX    | 251            | \$3.4B        | \$112.4M          | \$3.6B     | \$811.5M | 4.4x                                     |
| CO    | 38             | \$643.4M      | \$26.6M           | \$670M     | \$321.6M | 2.1x                                     |
| WI    | 71             | \$513.5M      | \$10.2M           | \$523.6M   | \$124.3M | 4.2x                                     |
| NE    | 93             | \$307.5M      | \$175M            | \$482.5M   | \$262M   | 1.8x                                     |
| AZ    | 11             | \$418.8M      | \$600.9K          | \$419.4M   | \$28.3M  | 14.8x                                    |
| MI    | 74             | \$379.5M      | \$6.8M            | \$386.3M   | \$55.5M  | 7x                                       |
| ND    | 53             | \$344.6M      | \$28.1M           | \$372.7M   | \$58.7M  | 6.4x                                     |
| MO    | 114            | \$368.4M      | \$2.1M            | \$370.5M   | \$117.9M | 3.1x                                     |
| KY    | 108            | \$321.6M      | \$3.4M            | \$325M     | \$62.4M  | 5.2x                                     |
| CA    | 8              | \$269.6M      | \$33.8M           | \$303.3M   | \$8.4M   | 36.2x                                    |

**Table 12.** Top 10 counties ranked by FEMA comparison (100-year return level)

| County             | Model                               | Hail Size  | Total Loss | FEMA EAL | FEMA Comparison (Total Loss vs FEMA EAL) |
|--------------------|-------------------------------------|------------|------------|----------|------------------------------------------|
| Cook Co., MN       | Stationary                          | 4.0+ in    | \$845.6K   | \$4.6K   | 184.1x                                   |
| Washington Co., ME | Nonstationary (Stationary fallback) | 4.0+ in    | \$2.1M     | \$18.3K  | 116x                                     |
| Emery Co., UT      | Stationary                          | 3.0-4.0 in | \$317.5K   | \$2.8K   | 113.9x                                   |
| Riverside Co., CA  | Stationary                          | 3.0-4.0 in | \$97.6M    | \$955.8K | 102.1x                                   |
| La Plata Co., CO   | Stationary                          | 3.0-4.0 in | \$1.4M     | \$14.2K  | 95.8x                                    |
| Carbon Co., UT     | Stationary                          | 3.0-4.0 in | \$485.9K   | \$5.5K   | 87.6x                                    |
| Chippewa Co., MI   | Stationary                          | 3.0-4.0 in | \$2.5M     | \$28.9K  | 87.3x                                    |

|                   |            |            |        |         |       |
|-------------------|------------|------------|--------|---------|-------|
| Lake Co.,<br>OR   | Stationary | 3.0-4.0 in | \$1.4M | \$16.1K | 85.5x |
| Luce Co.,<br>MI   | Stationary | 4.0+ in    | \$5.9M | \$71.2K | 83.2x |
| Navajo Co.,<br>AZ | Stationary | 3.0-4.0 in | \$4.9M | \$61.9K | 79.8x |

Figure 1.

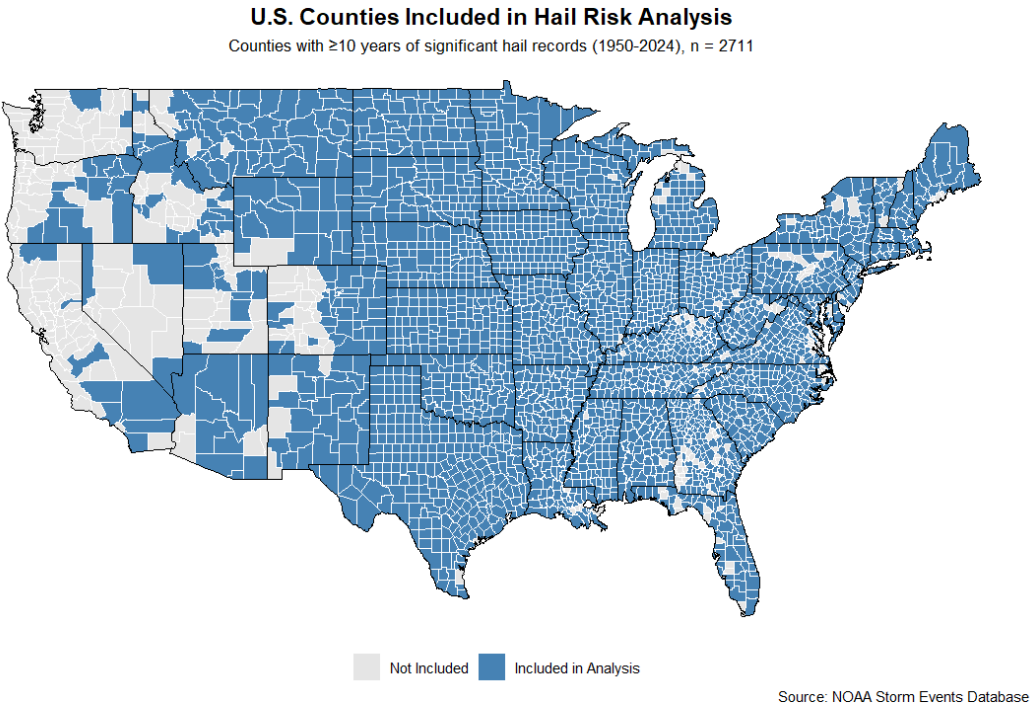


Figure 2.

Stationarity Assessment of U.S. County Hail Risk (1950-2024)

Stationary: 2380 | Nonstationary Increasing: 42 | Nonstationary Decreasing: 289

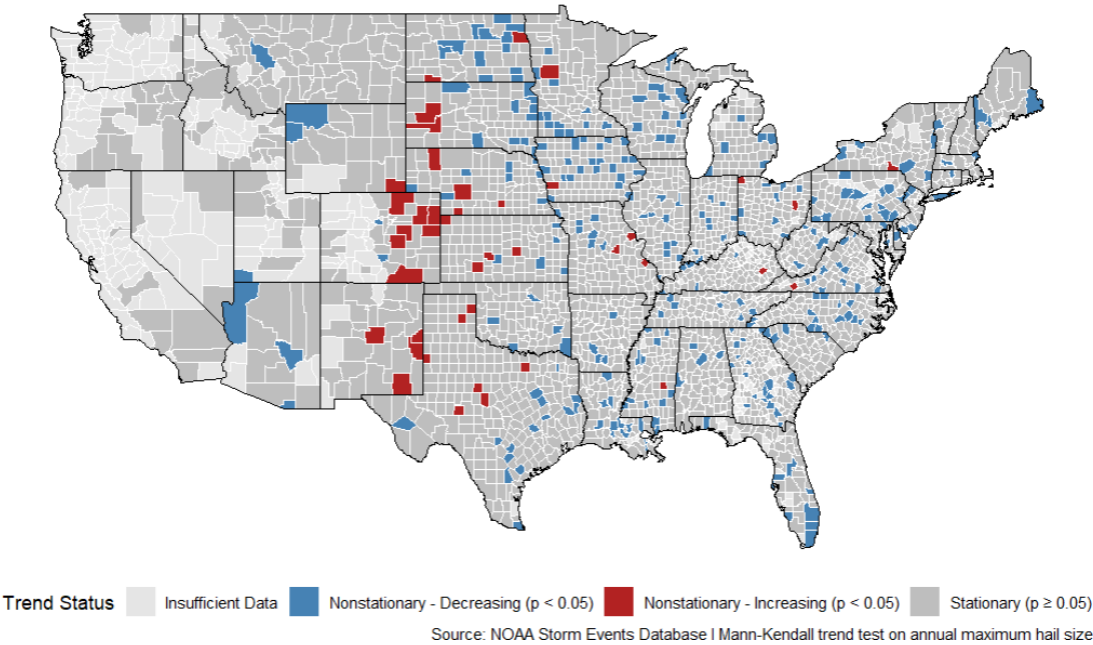


Figure 3.

Stationary GEV Parameter Distributions (L-moments)

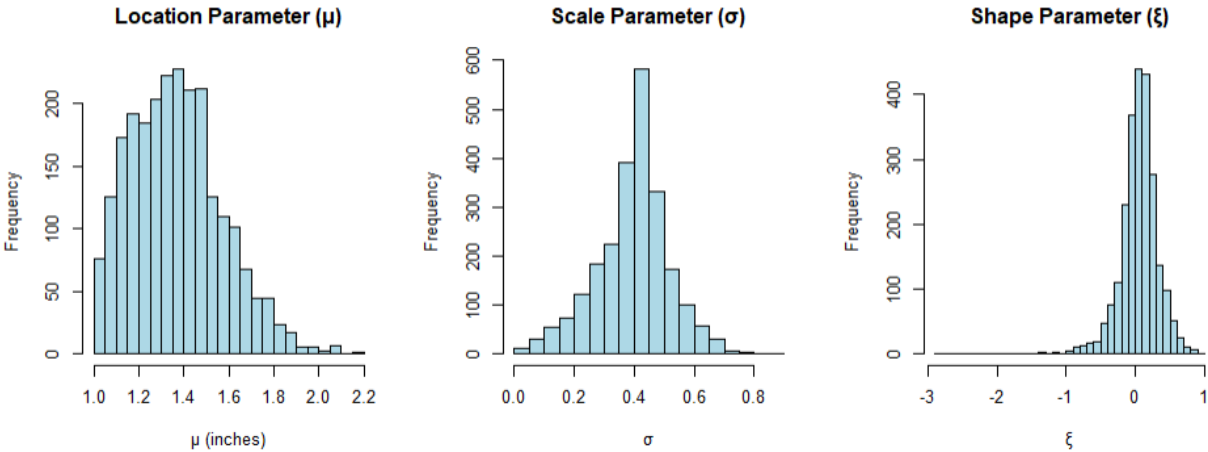


Figure 4.



### Stationary GEV Return Level Distributions (L-moments)

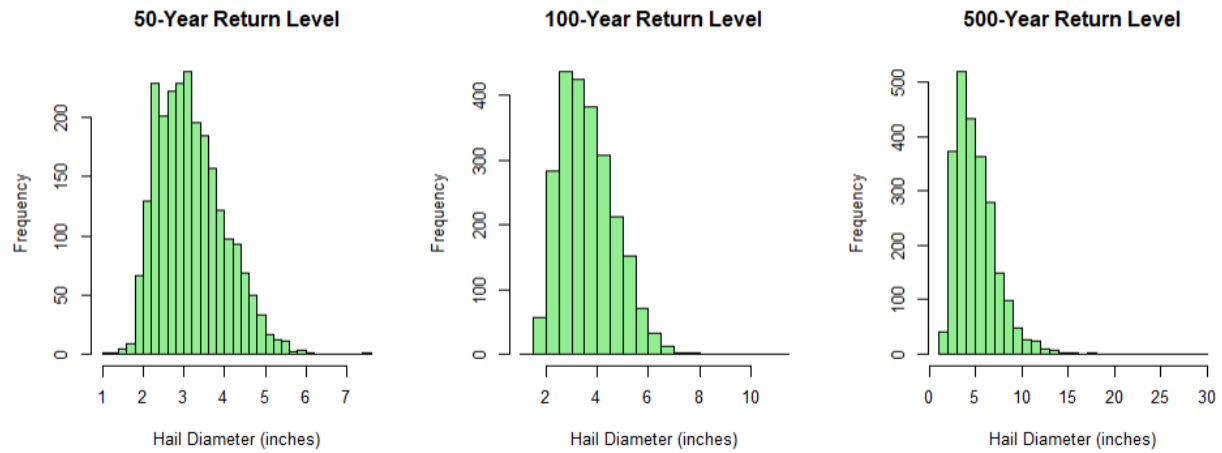


Figure 5.

### Nonstationary GEV Parameter Distributions (MLE)

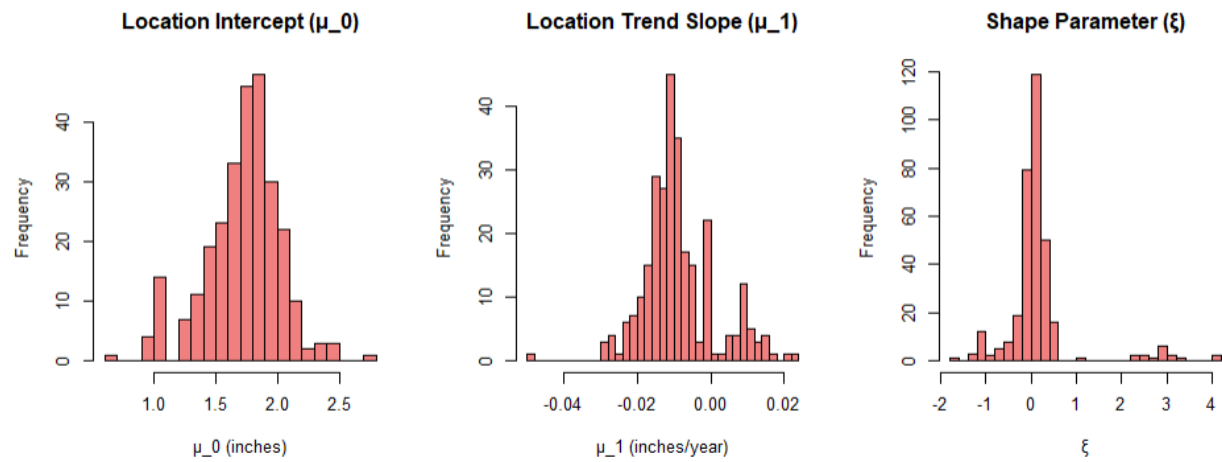


Figure 6.

Nonstationary GEV Return Level Distributions (MLE)

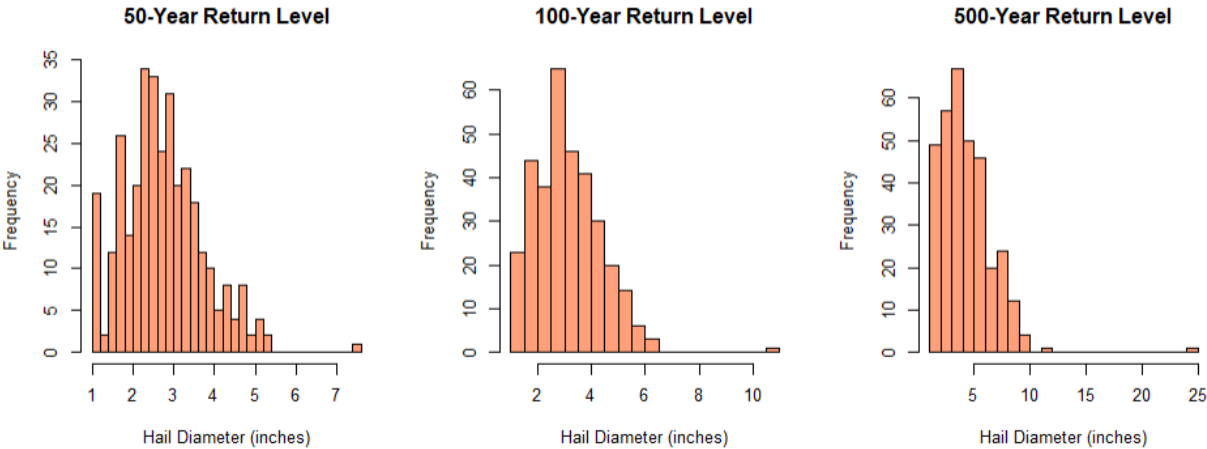
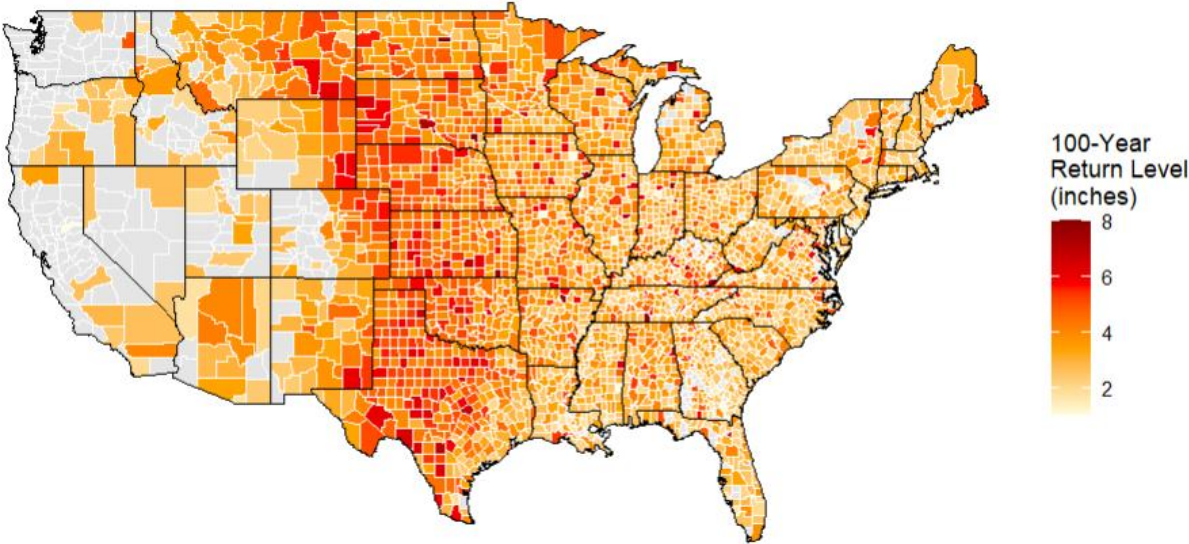


Figure 7.

100-Year Hail Return Level Across U.S. Counties

n = 2711 counties | L-Moments (stationary) and MLE (nonstationary)



Source: NOAA Storm Events Database (1950-2024)

Figure 8.

### Bootstrap Return Level Distributions - Oklahoma County, OK (Stationary)

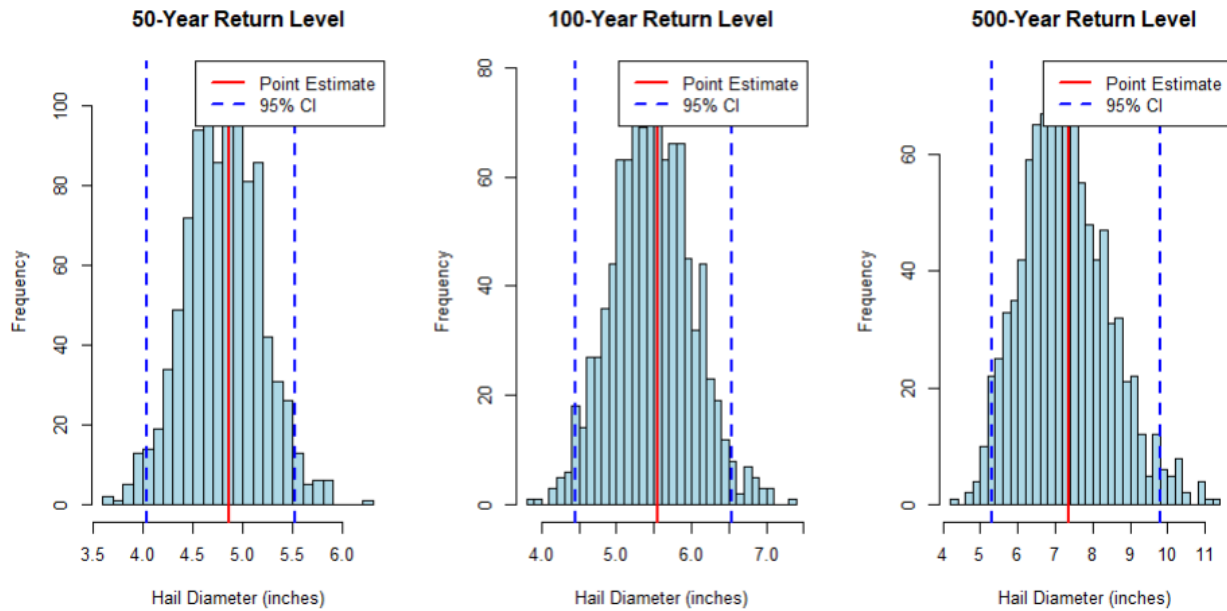


Figure 9.

### Bootstrap Return Level Distributions - Pennington County, SD (Nonstationary Increasing)

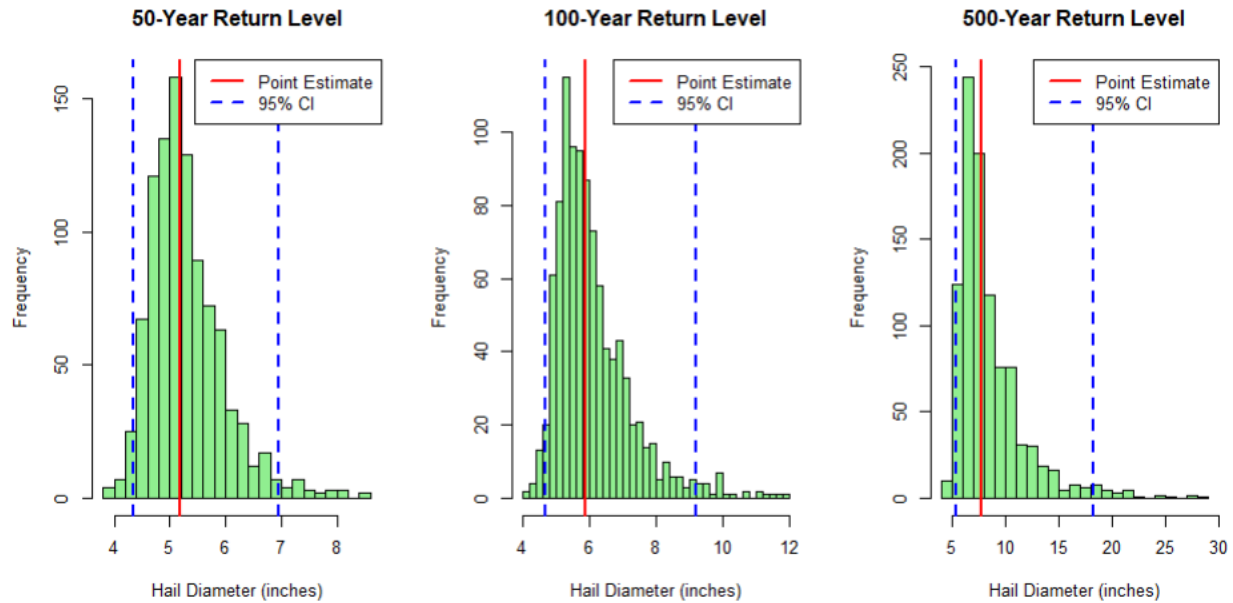


Figure 10.

### Bootstrap Return Level Distributions - Jefferson County, AL (Nonstationary Decreasing)

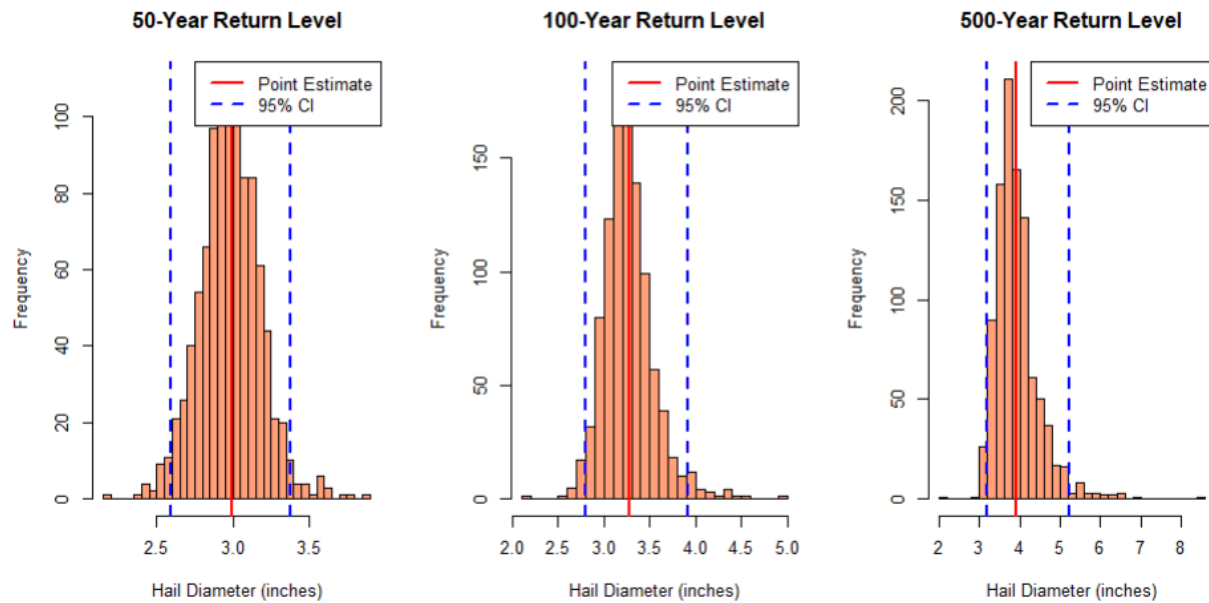


Figure 11.

### MCMC Return Level Plot - Oklahoma County, OK (Stationary)

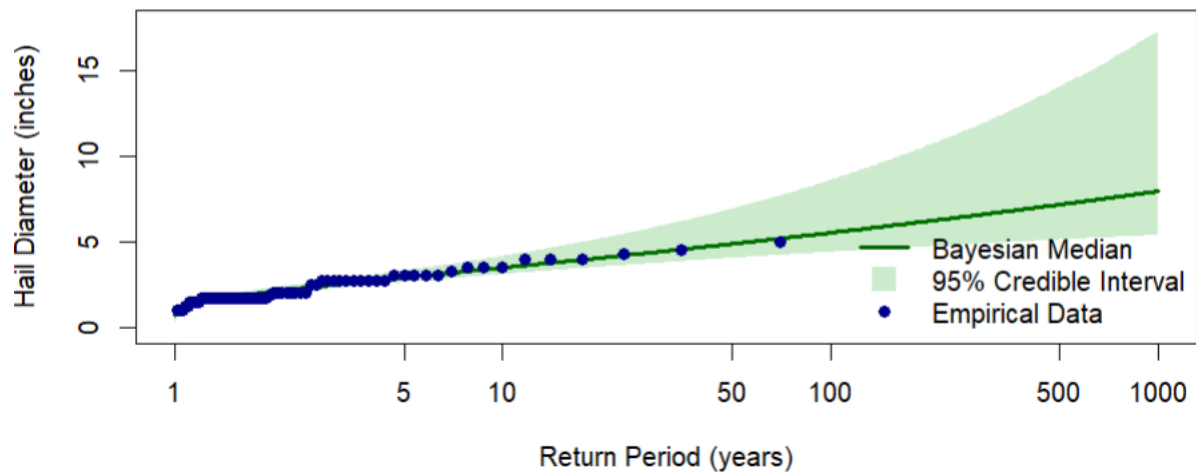
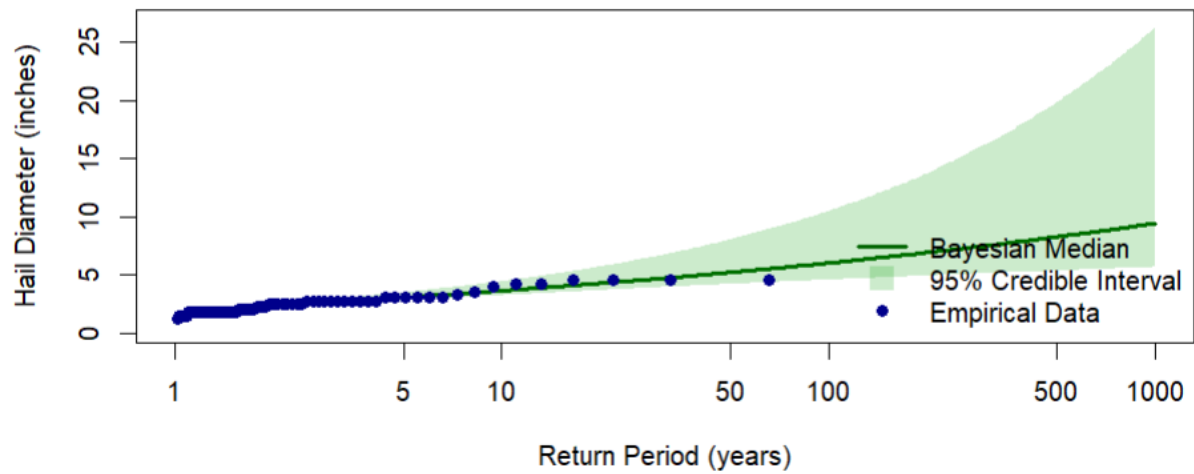


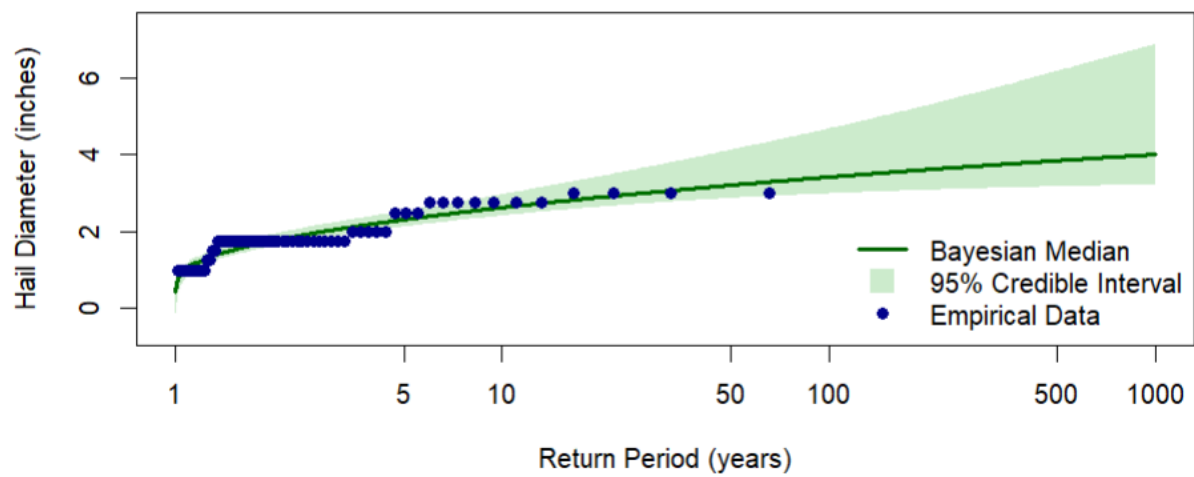
Figure 12.

**MCMC Return Level Plot - Pennington County, SD (Nonstationary Increasing)**



**Figure 13.**

**MCMC Return Level Plot - Jefferson County, AL (Nonstationary Decreasing)**



**Figure 14.**

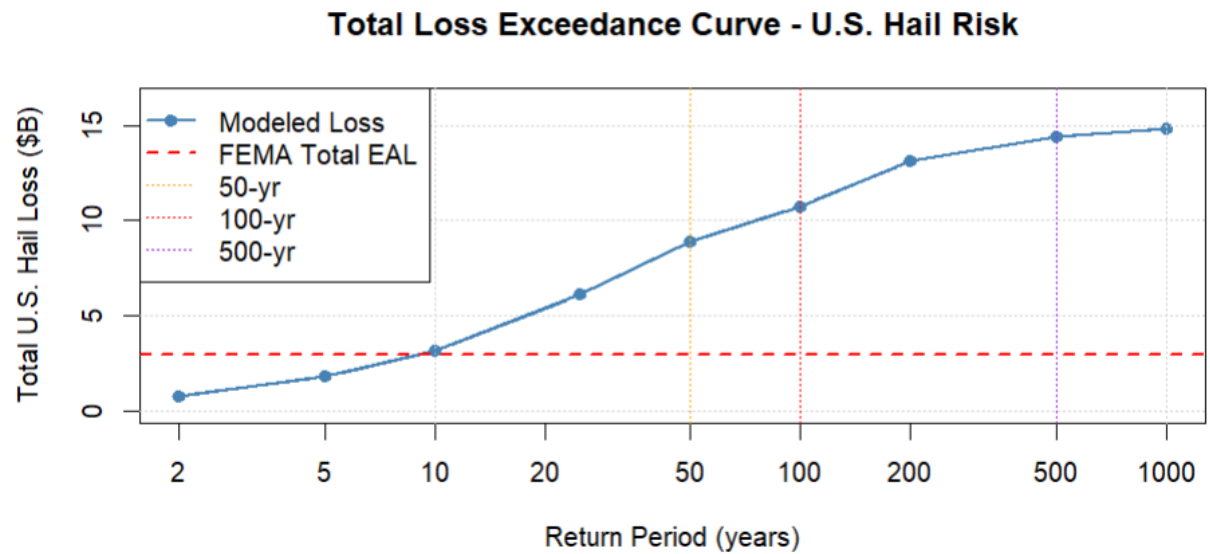


Figure 15.

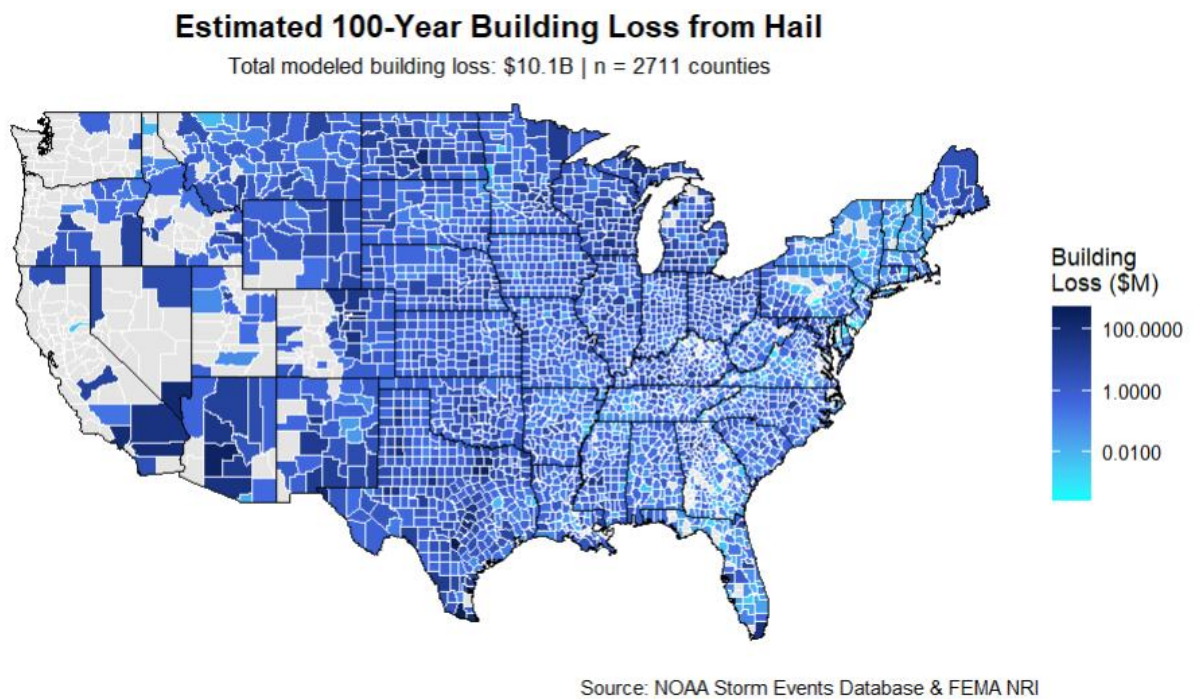
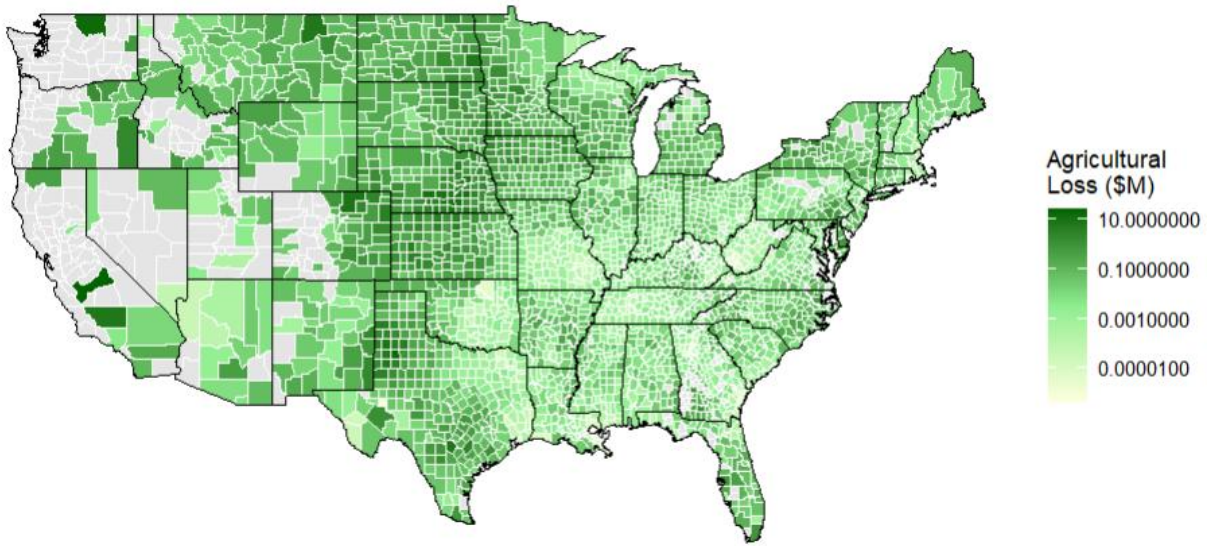


Figure 16.



### Estimated 100-Year Agricultural Loss from Hail

Total modeled agricultural loss: \$663.8M | n = 2711 counties

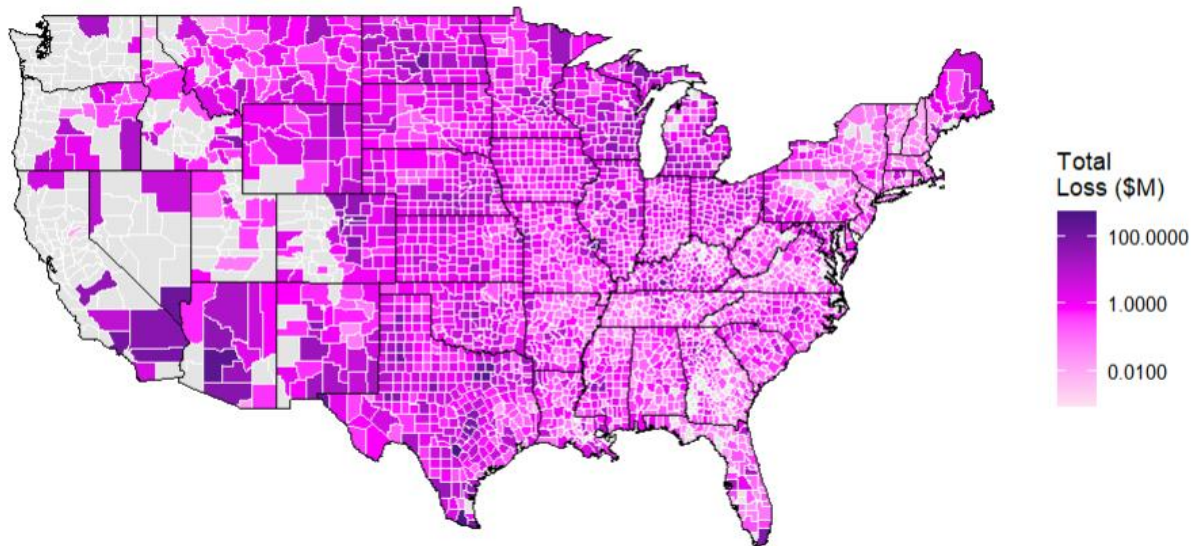


Source: NOAA Storm Events Database & FEMA NRI

Figure 17.

### Estimated 100-Year Total Hail Loss (Buildings + Agriculture)

Total modeled loss: \$10.8B | n = 2711 counties

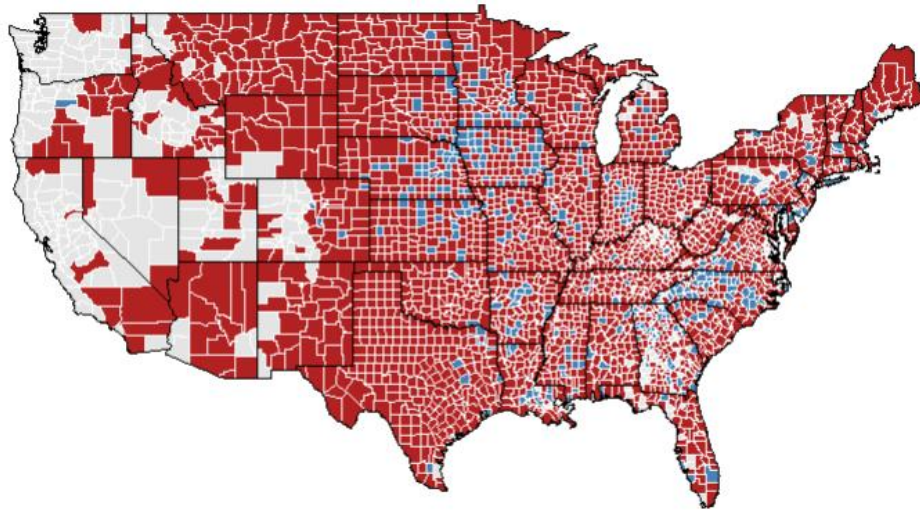


Source: NOAA Storm Events Database & FEMA NRI

Figure 18.

## Modeled 100-Year Loss vs FEMA Expected Annual Loss

Higher than FEMA: 2361 counties | Lower than FEMA: 350 counties



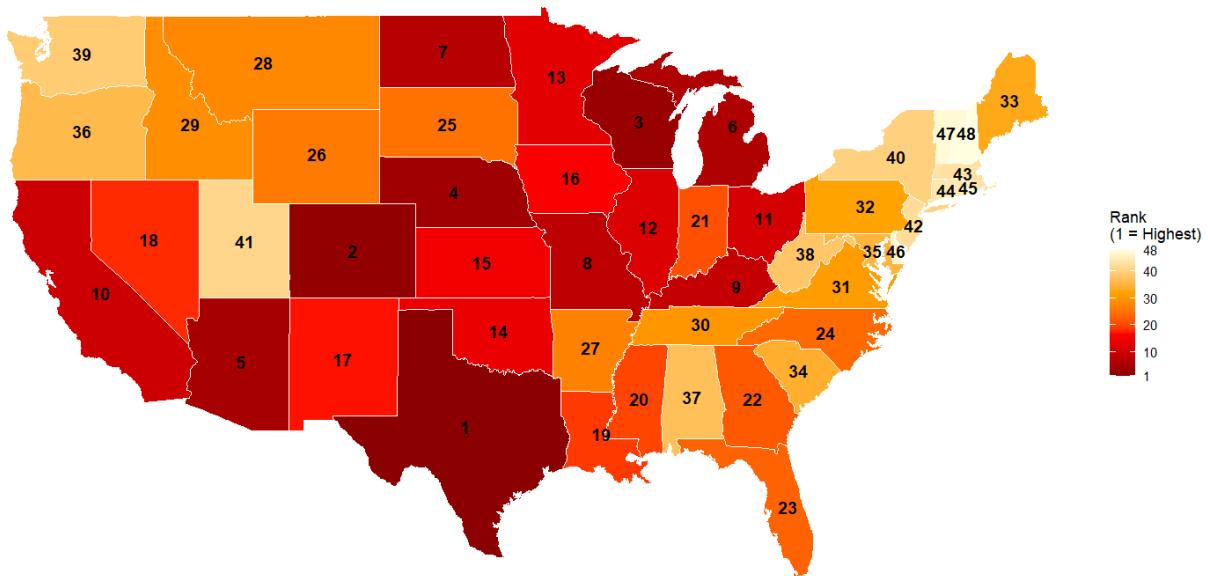
vs FEMA EAL ■ Higher than FEMA EAL ■ Lower than FEMA EAL ■ No Data

Source: NOAA Storm Events Database & FEMA NRI

Figure 19.

## U.S. State Hail Risk Ranking by 100-Year Total Loss (Lower 48)

Ranked by estimated building + agricultural loss at 100-year return period  
Total modeled U.S. loss: \$10.8B



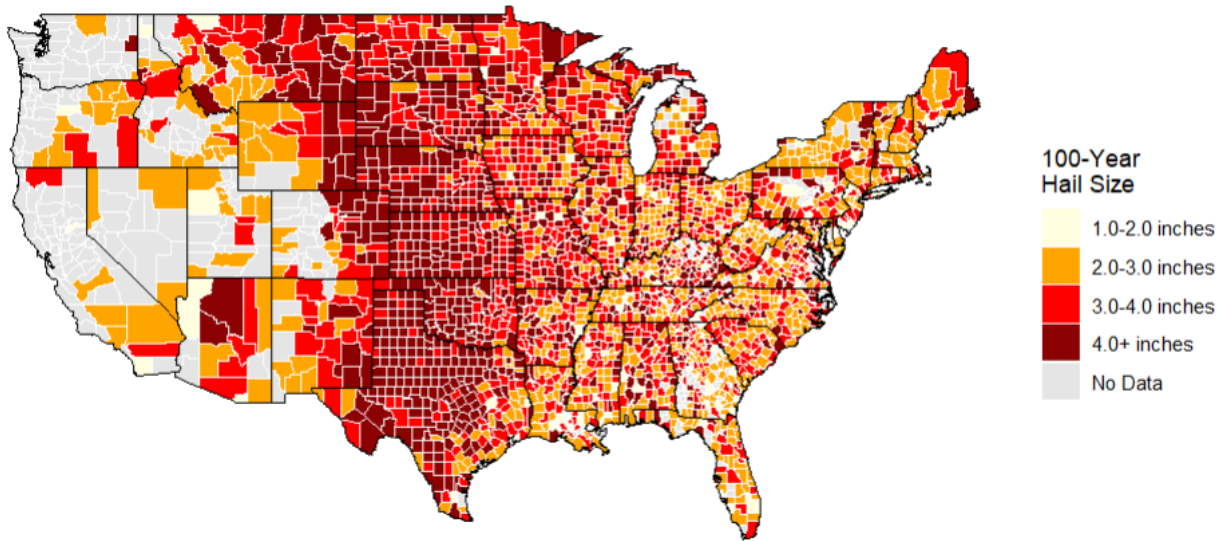
Source: NOAA Storm Events Database & FEMA NRI

Figure 20.



## 100-Year Hail Size Category Across U.S. Counties

Hail diameter bin classification based on GEV return level estimates |  $n = 2711$



Source: NOAA Storm Events Database (1950-2024)